

AI Fallibility & Automation Bias

Long-term Effects

Study of the Long-term Effect of Indicating the Fallibility of Artificial Intelligence on Automation Bias in Mental Health Application Users.

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* contributed equally to this study

N=24×2

Participants

1 Wk

duration

Background & Motivation

Mental Health and AI

Artificial Intelligence (AI) is increasingly used in mental health applications to support users and improve access to care. However, users may develop automation bias, meaning they overtrust AI recommendations without critically evaluating them. In mental health contexts, this can have serious consequences.

This Study

AI fallibility disclaimers have been proposed to reduce overreliance, but their long-term effectiveness is still unclear. This study examines how such disclaimers influence users' trust and evaluation behavior over time, contributing to the design of safer and more trustworthy AI systems.

Research Question

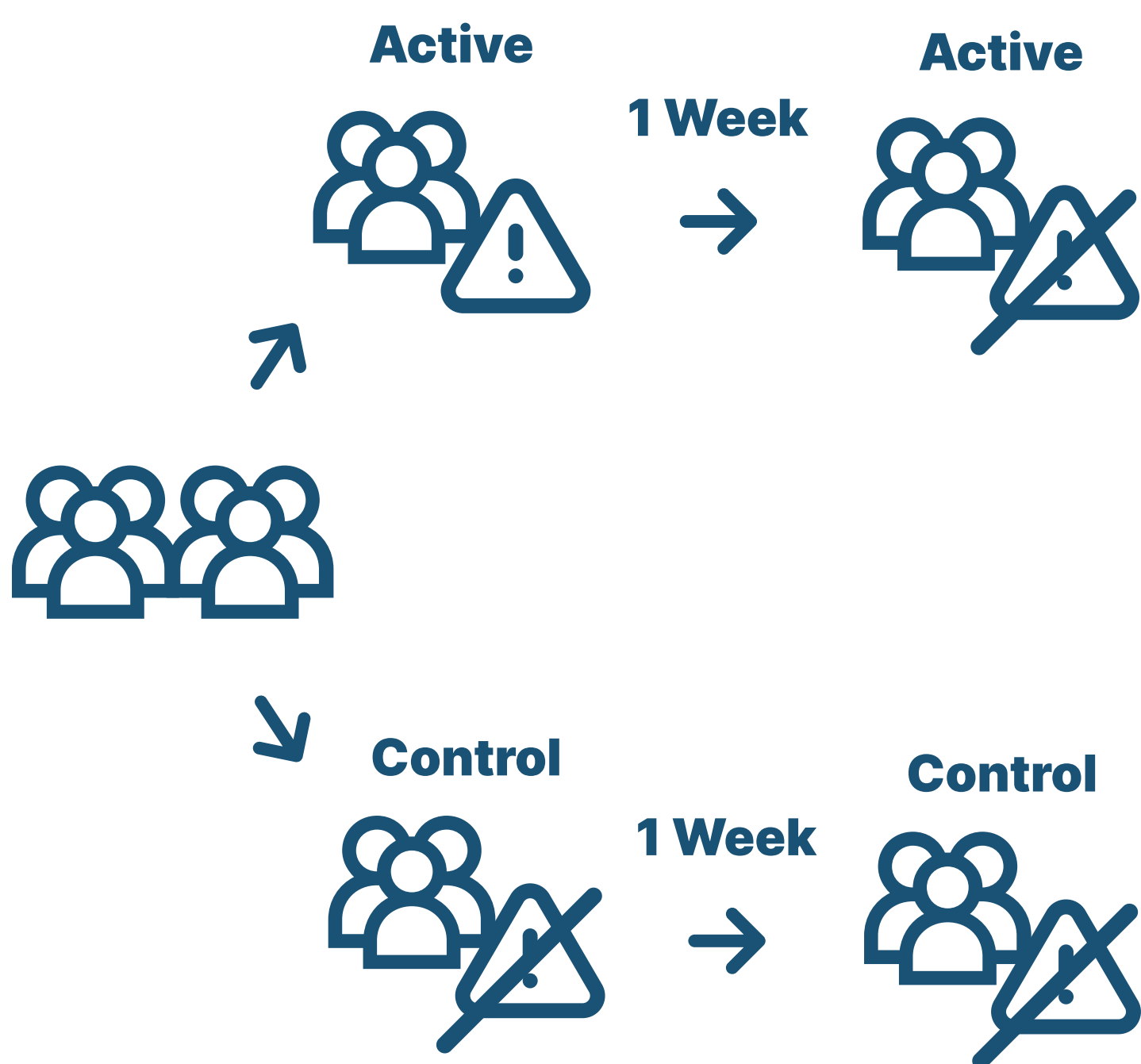
RQ

How effective are AI fallibility disclaimers in reducing automation bias and shaping users' trust in AI-supported mental health applications over time?

Study Design & Methodology

Study Design

The study was conducted as a mixed longitudinal experiment with two sessions separated by one week. The Participants were assigned to either the active group, which received AI fallibility disclaimers, or the control group, which did not receive disclaimers. The goal was to investigate short-term and long-term effects of disclaimers on automation bias and trust in AI recommendations.



Procedure

Each session consisted of 3 rounds. In each session participants completed a pre-study questionnaire about their attitudes toward AI. In a round participants read a mental health case vignette. After that they used an AI-supported mental health application to request an assessment. Then, the AI generated a treatment recommendation, and finally participants evaluated the recommendation using a post-interaction questionnaire to finish the round. After one week the procedure was repeated with both groups, but now both without disclaimers.

Measures & Instruments

Measures

In the study we measured automation bias through a rating of the AI answer and confidence in the rating. We have also included UEQ+ scales (Intuitive Use, Trustworthiness, Response Behaviour, Response Quality, Clarity, and Risk Handling).

Instruments

The patient vignettes were printed on paper, and for the measures we used a survey made with tally. The "AI" was presented in a PWA on a Phone.

Hypotheses

H₁

Exposure to a design intervention in a mental health AI application will lead to a substantial reduction in automation bias scores, such that participants in the intervention group will exhibit lower automation bias scores than participants in the control group.

H₂

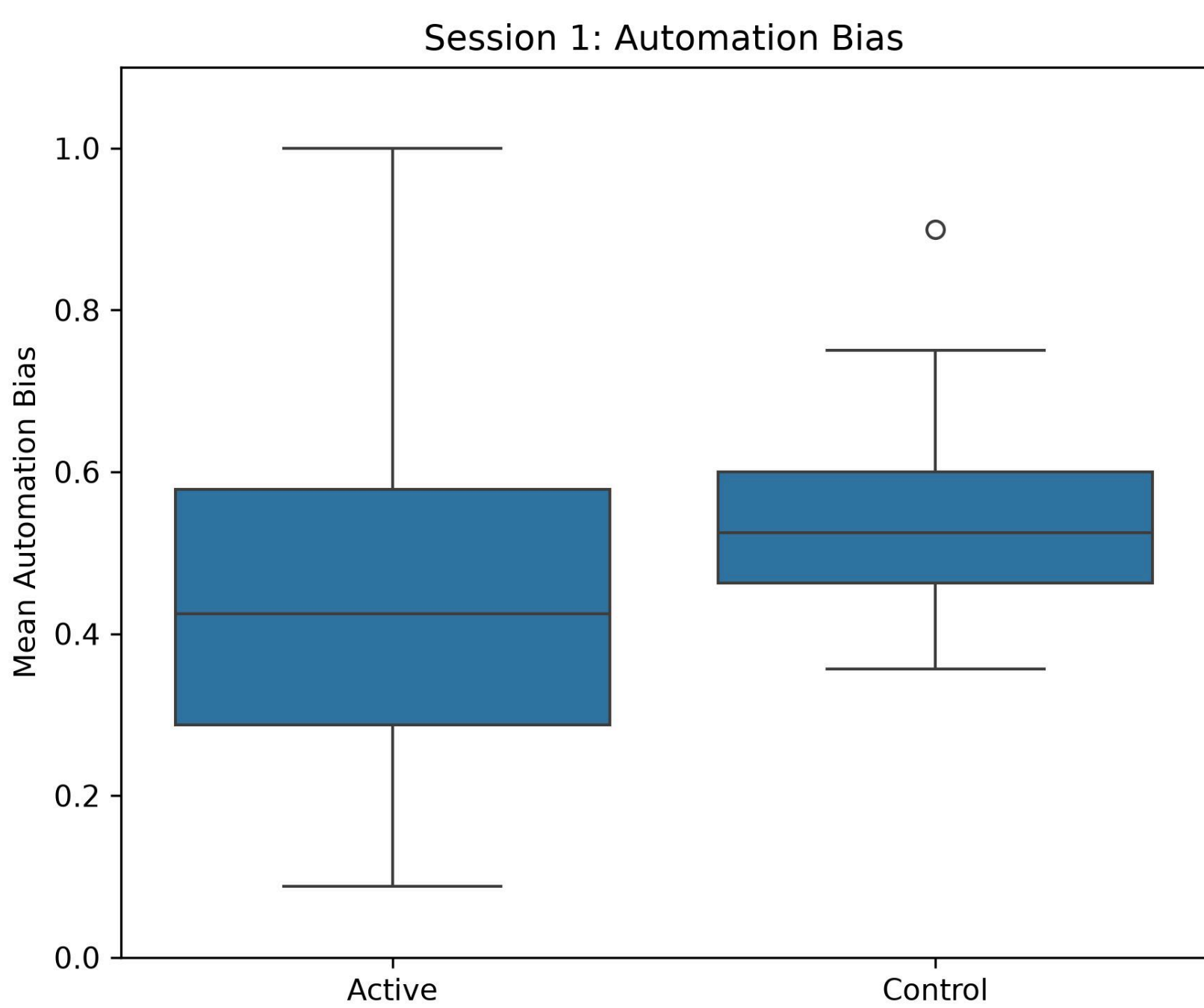
There will be a prolonged effect of indicating AI-fallibility, so that participants from the active group will exhibit lower automation bias scores even after a time gap of 1 week between session 1 and session 2, and without being shown the design intervention again.

Conclusion

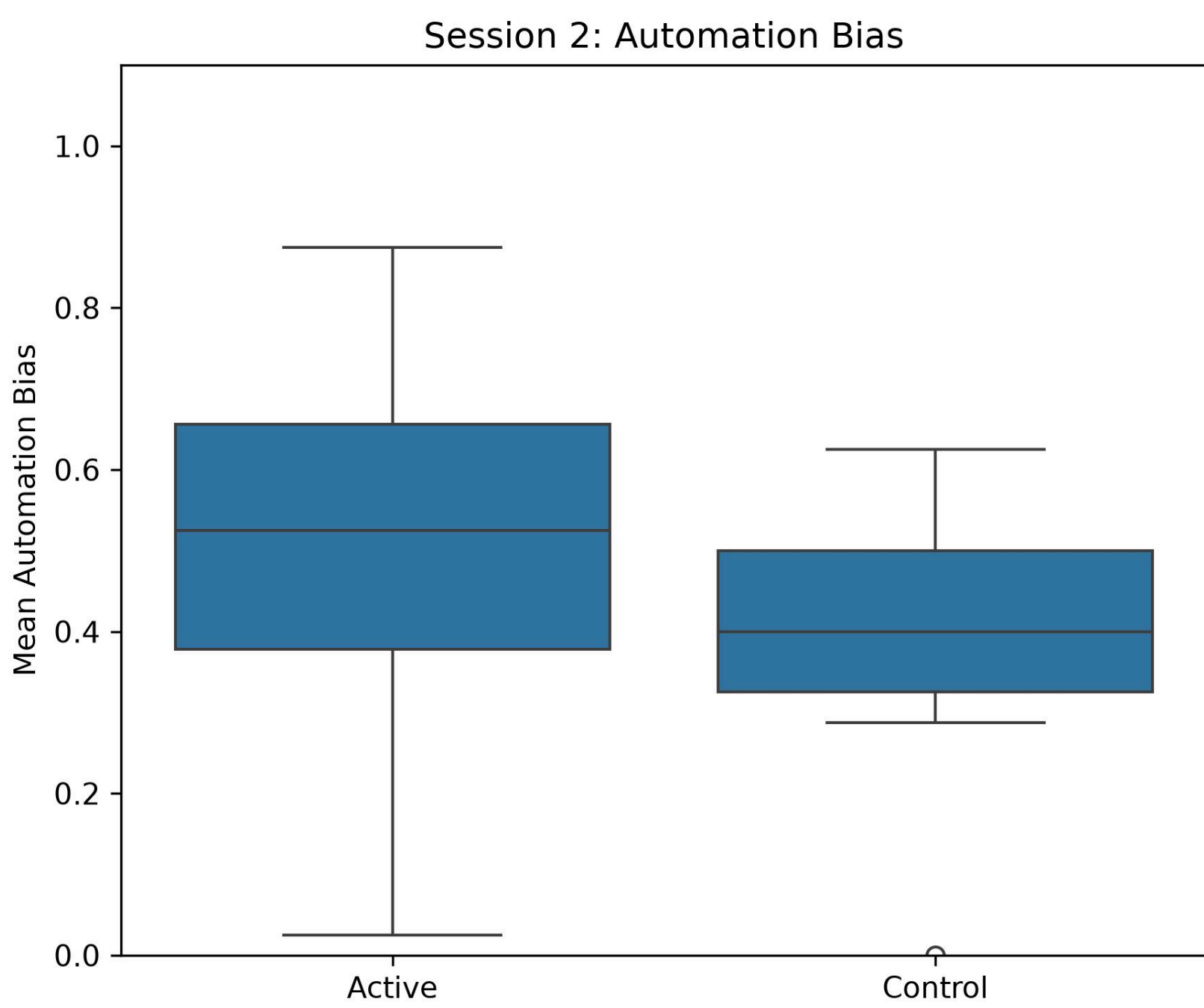
Conclusion

The automation bias results of the first session suggest that, at first, disclaimers can have a positive effect. The results from the second session do not suggest a longevity of this effect. Moreover, people who did not receive disclaimers in the first session exhibited better results in the second session compared to the first. Looking at the UEQ+ results, we can see that the results from the control group of the second session differ notably compared to the other rather coherent results.

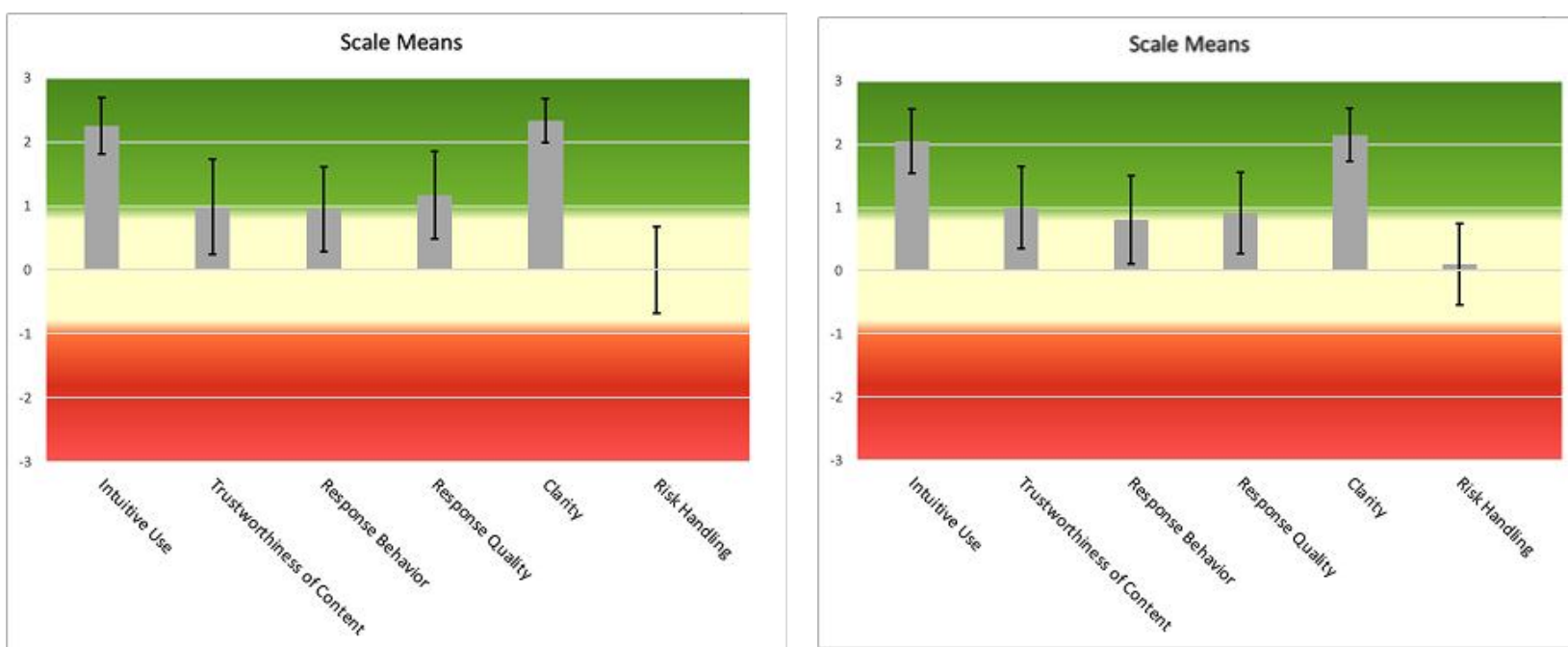
Results



Results suggest positive effect of disclaimers.

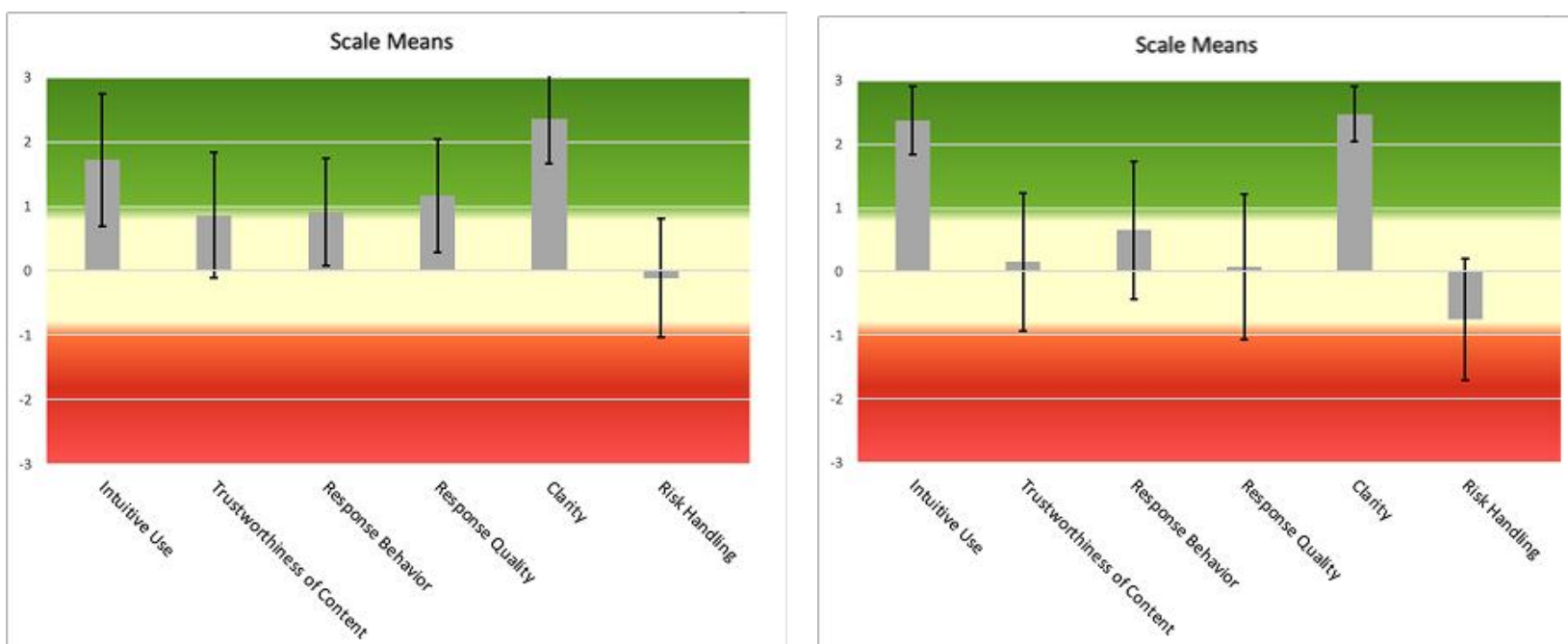


Results do not suggest positive long-term effect.



UEQ+ Active (Ses. 01)

UEQ+ Active (Ses. 02)



UEQ+ Control (Ses. 01)

UEQ+ Control (Ses. 02)

